Salah Boubnider University of Constantine 3
Faculty of Medicine
First Year of Medicine

PROTEINES

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Course Outline:

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II. Biological Importance
III. Protein Structure
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I. Introduction

Proteins are polymers (macromolecules) composed of one or more chains of amino acids (polypeptide chains). They are essential for both the structure and the functioning of cells, and it is through them that genetic information is expressed.

II. Biological Importance

Proteins are biomolecules of fundamental importance:

• **From a quantitative point of view**, they constitute 55 to 85% of the dry weight of the cell, making them the main component of the organism after water.

• **From a qualitative point of view**, they are involved in almost all cellular functions, with a wide variety of roles:

o **Structural role:** they provide support and protection for biological structures, such as collagen in connective tissue and cytoskeletal proteins (actin, tubulin), which are responsible for cell shape.

o **Biochemical catalytic role:** enzymes catalyze virtually all biochemical reactions.

Transport role: hemoglobin transports oxygen from the lungs to tissues and carbon dioxide in the opposite direction. Proteins also ensure the transport of molecules across cell membranes and regulate exchanges between the cell and the extracellular environment, such as Na^+/K^+ ATP-dependent pumps.

o **Regulatory role:** they participate in intra- and intercellular communication, coordinating metabolism within the cell and between different levels of the organism through hormones, their receptors, signaling proteins, and transcription factors.

o **Protective role:** immunoglobulins play a key role in immune defense.

o **Movement role:** actin and myosin are the proteins responsible for muscle contraction.

o **Role in energy production:** they can be involved when needed.

II. Protein Structure

Protein structure includes: primary structure, secondary structure, tertiary structure, and quaternary structure.

III.1. Primary Structure of Proteins

This refers to the linear sequence of amino acids (AAs), that is, the order in which amino acids are linked within the polypeptide chain. Peptide bonds therefore form the basis of the primary structure, but they are not the only types of bonds involved. Indeed, proteins may be composed of several different polypeptide chains. These chains are held together by secondary interactions, particularly hydrogen bonds and disulfide (S–S) bridges. These disulfide bonds cause distortions in the polypeptide chains.

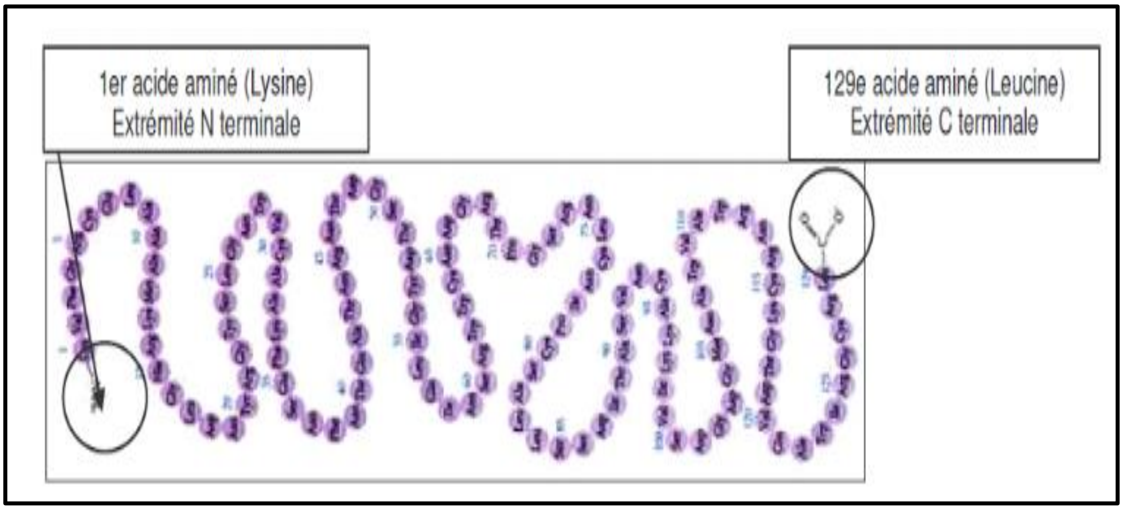

Figure 1: Primary structure of proteins

The protein does not retain this initial structure and very rapidly undergoes a three-dimensional evolution into secondary, tertiary, and quaternary structures. This three-dimensional evolution ultimately leads to a well-defined spatial conformation: the native protein.

Note: In biochemistry, the native state of a protein is its functional or operative form.

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III.3. Secondary Structure

This corresponds to a regular arrangement of amino acids along an axis. Two main types of secondary structure are distinguished:

- **α-helix**
- **β-sheet**

α-Helix

The α-helix is formed by the regular rotation of a polypeptide chain around itself (in a spiral) to the right (clockwise). The helix is stabilized by hydrogen bonds between the C=O and N-H groups that are four residues apart in the primary sequence.

- The hydrogen bonds run parallel to the axis of the helix.
- The R side chains are oriented outward from the helix.
- **Example:** Hair keratin is a fibrous protein that exists as an α-helix along its entire length.

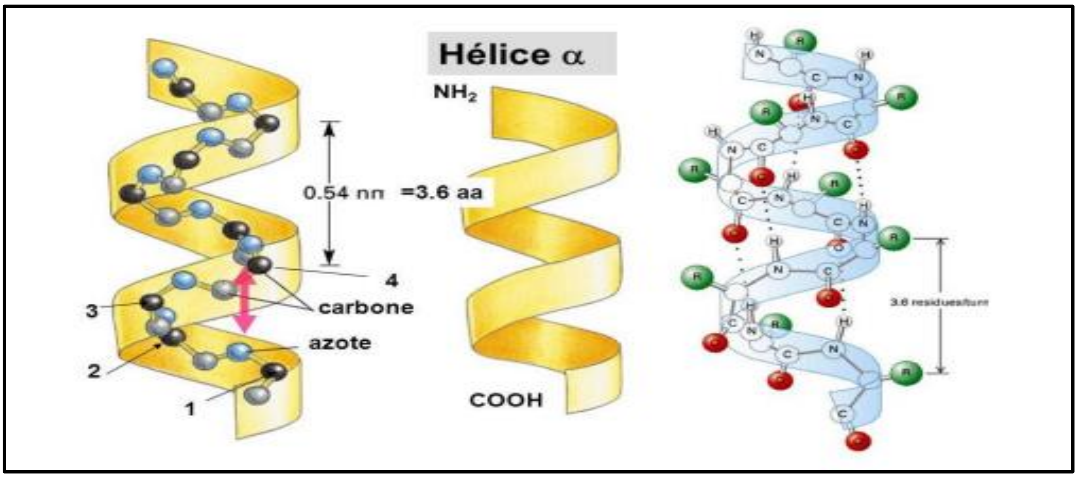

Figure 2: Structure of the α-helix

β-Sheet

The β-pleated sheet differs significantly from the α-helix in that it is an almost fully extended sheet (zig-zag structure) rather than a rod-like (helical) structure.

- The side chains alternate, pointing upward and downward.
- Hydrogen bonds are perpendicular to the axis of the polypeptide chain.
- β-sheets can be:

  ¬ **Parallel** when the chains run in the same direction.

  ¬ **Antiparallel** when the chains run in opposite directions (more stable due to less distortion).

Example: Fibroin is a protein secreted by the silkworm that forms silk threads and is primarily composed of β-pleated sheets.

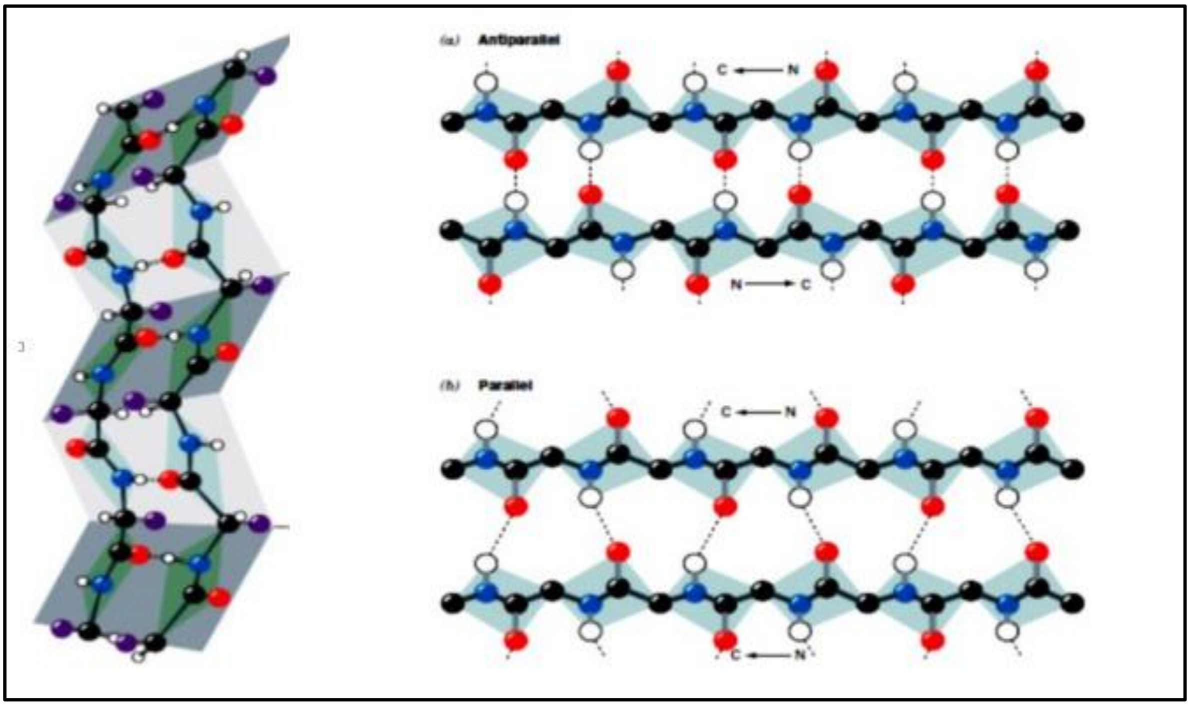

Figure 3: Structure of the β-sheet

Turns and Loops

Outside of α-helices or β-sheets, short amino acid sequences adopt irregular structures forming **turns** (Figure 4).

- **Turns:** 2 to 4 amino acids connecting two antiparallel β-sheets.
- **Loops:** Generally about ten residues long, connecting β-sheets or a β-sheet and an α-helix. They are usually located on the protein surface.
- Glycine and proline residues are frequently found in these regions.

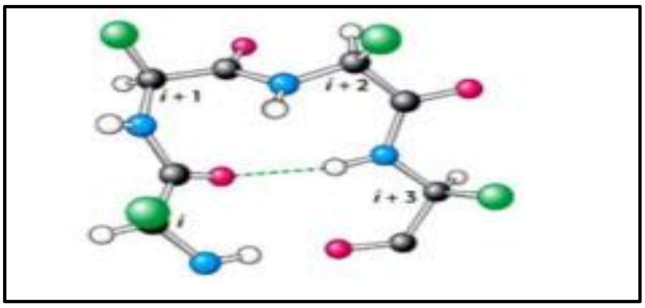

Figure 4: Illustration of a Turn

III.3. Tertiary Structure

The tertiary structure is defined by the folding of the polypeptide chain, organized into elementary α or β units, into a complex three-dimensional configuration. The tertiary structure mainly arises from interactions between the R groups of the amino acids composing the protein.

- It is stabilized by various non-covalent bonds (hydrogen, ionic, hydrophobic, Van der Waals) and sometimes by covalent bonds, such as disulfide bridges.
- It is flexible and can change (twist, deform) in response to ligand binding or variations in physicochemical parameters (pH, temperature).

Example: Myoglobin and ribonuclease.

Types of bonds:
The bonds that can form between the amino acids making up a protein can be classified as:

- **Covalent bonds:** Peptide bond, disulfide bridge.
- **Non-covalent bonds (or weak bonds):** Hydrogen bond, hydrophobic interaction, ionic bond, Van der Waals forces (or Van der Waals bond).

1- Hydrogen bond: Forms when a hydrogen atom bonded to an electronegative atom (most often nitrogen) is near another electronegative atom (most often oxygen).

2- Hydrophobic interaction: The side chains of nonpolar (apolar) amino acids associate with each other in the interior of the protein to avoid interacting with water, while the more polar residues remain in contact with the solvent, contributing to the protein's compact folding.

3- Ionic bond: It forms between ions with opposite charges.

4- Van der Waals interaction: These correspond to a weak electrical interaction that occurs at short distances between atoms and/or molecules. They allow atoms to position themselves as close as possible to each other, enabling a very compact structure in the protein's core.

III.4. Quaternary Structure

Some proteins exhibit a higher level of organization, known as the quaternary structure. In this case, these proteins are composed of several polypeptide chains called subunits or protomers.

They are described as **homopolymeric** (identical subunits) or **heteropolymeric** (different subunits).

- The subunits do not assemble randomly; the whole structure has symmetry.
- They are connected according to a specific geometry by **weak bonds** (hydrogen bonds, hydrophobic interactions, ionic bonds, Van der Waals interactions), except for disulfide bridges.

Example: Hemoglobin (Hb).

In adults, Hemoglobin A (95%) consists of a protein portion made up of 4 polypeptide chains ($\alpha_2\beta_2$) and 4 heme prosthetic groups (non-protein portion), in which the iron atoms are all in the ferrous form Fe^{2+} (see Figure 5).

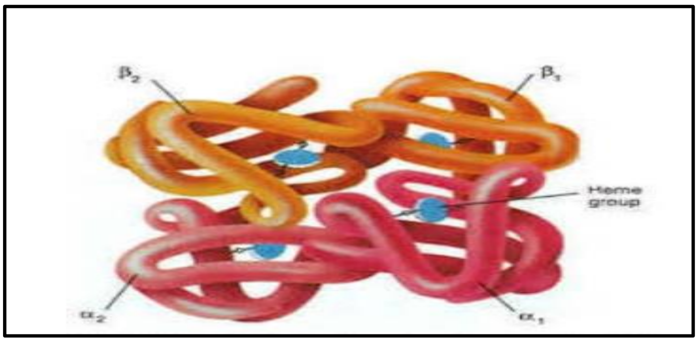

Figure 5: Quaternary structure of hemoglobin.

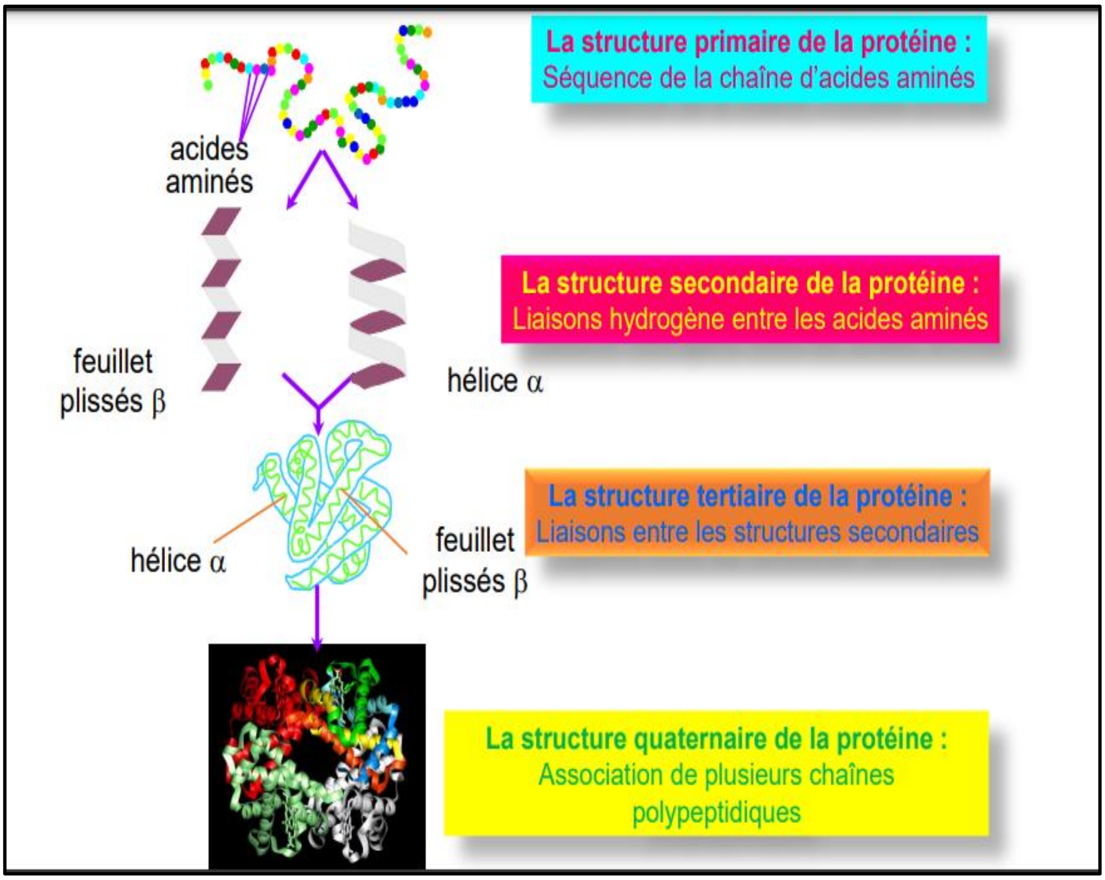

Figure 6: Structural evolution of a protein

Note:

1. Not all proteins develop a quaternary structure.
2. **Thermal stability of proteins:** A significant increase in temperature above 40°C or exposure to cold can induce denaturation (loss of the protein's biological activity).

- **Denaturation:** Modification of the three-dimensional structure without altering the primary structure

VI. Classification of Proteins

VI.1. According to their composition: There are:

a) **Holoproteins:** Entirely composed of amino acids.

- **Structural proteins:** Collagen
- **Muscle proteins:** Actin, myosin
- **Soluble proteins:** Albumin, globulin

b) **Heteroproteins:** Composed of a protein portion called apoprotein and a non-protein portion that contains:

- A **prosthetic group** (mineral, metallic, or organic).
- A **carbohydrate**: glycoproteins
- A **lipid**: lipoproteins
- A **nucleic acid**: nucleoproteins

VI.2. According to shape:

- **Globular proteins:** Globular proteins play an essential role in metabolism and perform many other functions, such as albumin, globulin, and histones (composed of α-helices and β-sheets connected by β-turns).
- **Fibrous proteins:** Fibrous proteins are found in tissues and structural elements such as muscles, bones, skin, and cellular components or membranes. They perform structural roles and are composed of units with simple, repetitive secondary structures (α-helix and β-sheet). Collagen and keratin are fibrous proteins with an α-helix conformation.

References

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